

Paper Manager pack — DROID (Nov 4 · Data)

Paper: DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset

arXiv: <https://arxiv.org/abs/2403.12945> · [project](#) · CC-BY 4.0

PDF: /workspace/papers/droid-2403.12945.pdf (= /workspace/droid-quiet/paper.pdf)

Session: Nov 4 · due 9:30 AM · Type: **Data**

Role weights: **Stakeholder=5 (LONGEST)** · **Scientific Reviewer=2 (substantial)** · Empiricist / Archaeologist / Visionary also filled (seats otherwise empty)

Kanta role: unspecified this paper — weight Stakeholder heaviest; Reviewer substantial; produce all five

Nearby pressure: **Nov 6 Midterm** — keep exam bites / exact numbers ready

Sources: Summarizer + Helper ablations + Senior Research (/workspace/droid-quiet/DELIVERABLE.md) + paper_notes.md / env_outline/

Figure IDs (paper-correct): **Fig. 8** = main co-train bars (+22% ID / +17% OOD); **Fig. 10** = matched scene-diversity ablation; **Fig. 6** = 3D interaction locations; **Fig. 9** = Cook Lentils qualitative. (Summarizer sometimes swapped Fig. 6/8 — use these.)

1. One-liner

Same-robot many-real-scenes + protocolized diversity beats shallower multi-embodiment pools for Franka OOD — steal DROID's *protocol* into BEHAVIOR, don't just drink another river.

Contrast with OXE: OXE = many robots / aggregated lab pools; DROID = **same Franka + Robotiq stack, 564 authentic scenes / 52 buildings**, GUI-forced task sampling + scene churn.

2. Summary (exec overview + key numbers)

DROID (Distributed Robot Interaction Dataset) is a distributed **in-the-wild** Franka teleop corpus: **76k** successful trajectories (~350 hours) across **564 scenes, 86 tasks (verbs), 52 buildings**, gathered by **50 collectors** at **13 labs** on **18** identical robot replicas over **12 months** (NA / Asia / Europe). ~16k fails are released but excluded from the paper's 76k count and co-train mixes.

Not a new foundation model. Experiments are language-conditioned **Diffusion Policies** (Robomimic) co-trained **50/50** with small in-domain demos. Aggregate vs next-best: **+22% absolute success in-distribution** and **+17% OOD** across **6 tasks × 4 locations (Fig. 8)**. Scene-diversity ablation at matched **7362 traj (Fig. 10)** shows **diversity—not just hours**—drives OOD. Co-train pool in paper = first ~40k language-ready successes (headline **76k** ≠ always train size).

Axis	Number
Successful traj / hours	76k / 350 h
Scenes / buildings	564 / 52
Verbs (tasks) / collectors	86 / 50
Institutions / robots	13 / 18
Control rate	15 Hz Polymetis
Policy horizons	obs 2 / pred 16 / action 8
Co-train lift (Fig. 8)	+22% ID / +17% OOD

Diversity ablate (Fig. 10)	7.3k diverse > 7.3k top-20 scenes (OOD)
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Motion FLAG: chunked reactive **Diffusion Policy** (obs=2 / pred=16 / action=8 @15 Hz) — **NOT** a classical motion planner / MPC / skill graph.

3. Keywords

DROID; in-the-wild manipulation; distributed collection; Franka Panda; Robotiq; ZED stereo; Quest teleop; scene diversity; Diffusion Policy; action chunking; receding horizon; co-training; OXE contrast; BridgeData V2; camera calibration; Polymetis; CC-BY 4.0; BEHAVIOR diversity protocol; π_0 / FAST / pi05_droid lineage

4. Deep dive — Motion planning & long-horizon control (PRIORITY)

Callout: Paper policies are **chunked reactive BC**, not classical planners.

Knob	Paper choice	Implication
Family	Diffusion Policy (DDPM/DDIM over action trajectories)	Denoising imitation, Robomimic backbone
Obs horizon	2	Short visual/proprio history
Prediction horizon	16 @ 15 Hz $\approx$ ~1.07 s planned	Chunk length
Action / exec horizon	8 open-loop then replan	Receding-horizon / action-chunking
Commit window	commit_s = 8/15 $\approx$ 0.53 s	Log next to SR (π_0 / OpenPI rhyme)
Action space (policy)	Absolute EE + gripper	Joints logged but not main policy space
Cams in policy	Two exterior RGB @ 128 ²	Wrist collected but underused in paper recipe
Vision / lang	ResNet-50 + frozen DistilBERT	Pre-VLA stack — pair data with modern heads in 2026

Long-horizon stress tests

Task	Class	In-domain demos	OOD	Why it matters
Close Waffle / Place Chips	Short	70 / 50	Distractors / novel	Baseline reactive
Apple in Pot / Toasting	Medium	60 / 150	Distractor plate / novel toast	Multi-object
Clean up Desk	Long	50	Desk / drawer distractors	Stage completion
Cook Lentils	Long	50	Distractors + camera shift	3-stage kitchen; Fig. 9 qualitative win

Cook Lentils requires **all 3 stages** (pour → pan → stove). DROID co-train finishes stages more reliably under OOD; No Co-train / OXE struggle after 1–2 stages (**Fig. 9**). Gains attributed to **diverse co-train data + chunked BC**, not hierarchical planning.

Syllabus placement: When contrasting “planner vs reactive policy,” park DROID firmly in **reactive chunked imitation** (same family as Octo / π chunk executors; contrast OpenVLA single-step AR). Horizon **16/8 @ 15 Hz** is the numeric tattoo next to Octo/ π_0 chunks.

Empiricist hook: Sweep pred $\in \{8,16,32\} \times$ exec $\in \{4,8\}$ on a multi-stage proxy; log commit_s. Paper recipe is **not** the 2026 BEHAVIOR controller — steal horizons literacy, ship $\pi_{0.5}$ / N1.7 heads.

5. Deep dive — Collection methodology & diversity (PRIMARY CONTRIBUTION)

FLAG: Methodology + diversity protocol is the paper’s main scientific product; policy results are the stress test.

Hardware (identical × 18)

Franka Emika Panda 7-DoF + Robotiq 2F-85; height-adjustable wheeled desk; 2× ZED 2 exterior + ZED Mini wrist; Quest 2 → continuous 6D + gripper; Polymetis **15 Hz**; joint **and** EE logged; checkerboard extrinsics at scene start (+ App. G post-hoc: ~**36k** cam → base, all-scene cam ↔ cam, curated ~**24k** quality set).

Protocol knobs (steal these)

- 1. **Move the robot, not just the objects** — ≤~**100 traj** / ~**20 min** per scene → forced scene turnover.
- 2. **GUI random task sampling** from collector-authored scene list → reduces easy-task bias.
- 3. **Periodic scene augmentations** — nudge base, move/recalibrate cams, lighting, add/remove clutter.
- 4. **Crowdsourced multi-instruction** post-label (≤**3** / episode, tasq.ai).
- 5. **Success flagging** at collect time; fails retained (~16k) but excluded from paper co-train.
- 6. **Shared portable stack** + public hardware guide → community multiplier.

Diversity axes (§IV)

Axis	DROID	Contrast
Scenes	564 / 52 buildings	BridgeV2 24 ; OXE agg ~ 311 ; RT-1 2
Viewpoints	1417 unique 3rd-person	Cams remounted / nudged
Interaction loci	Wider 3D first-grasp workspace (Fig. 6)	Tabletop-fixed priors
Verbs / objects	86 long-tailed; everyday categories	Only BridgeV2 comparable verb tail
Embodiment	One public Franka stack	OXE = 22 embodiments

Causal-ish evidence

- **Fig. 8 (mix):** No Co-train vs OXE 50/50 vs DROID 50/50 → DROID **+22% ID** / **+17% OOD** vs next-best (10 rollouts/cell; 6 tasks). Co-train uses first **40k** lang-ready successes; OXE baseline ≈ Octo split minus Language Table (~**400k**).
- **Fig. 10 (diversity @ fixed N):** **7362** from top-20 scenes vs **7362** diverse → diverse **higher OOD** (avg bars ~0.40 vs ~0.60); full DROID ≥ subsampled diverse on overlapping tasks.

Moral: Composition of scenes beats raw demo count; OXE breadth ≠ substitute for authentic scene entropy on the eval embodiment.

6. Role-ready sections (all five)

6.1 Stakeholder — WEIGHT 5 / LONGEST

Product one-liner: Open **Franka in-the-wild** flywheel — **76k / 350 h / 564 scenes** — co-train with a handful of site demos to buy ~**+20%** robustness **without** buying 22 embodiments.

Who pays / who benefits

- **Franka labs:** clone cart (hardware guide), add **50–150** local demos, co-train 50/50 before inventing architecture.
- **Foundation teams (OXE / OpenVLA / π / OpenPI):** DROID = open **scene-diversity** shard *and* mix-bug **canary** (OpenVLA drop).
- **BEHAVIOR’26 organizers / entrants:** proof that **scene count × protocol** moves OOD even when traj count < OXE / RT-1.

- **Non-consumers:** no Franka/ZED budget; need native bimanual/mobile-base data (DROID is single-arm desk/room-centric).
- Decision framing (Nov 4 → midterm Nov 6)**
1. **Have Franka + a week of teleop?** Collect 50–150 demos → paper DP recipe (or modern chunk head) **before** new architecture. Midterm answer: cite **+22/+17** and **564 scenes**.
 2. **Mixing into a VLA?** Treat DROID as a **monitored shard** — watch action loss; pre-define **kill switch** (OpenVLA loyalty lesson).
 3. **Designing BEHAVIOR data?** Copy **protocol** (random prompts, scene time-caps, view/lighting jitter, success flags) — not only vibes / another download.
 4. **Budget truth:** 13 institutions × 12 months × 50 collectors is the hidden cost; open release socializes it. One building ≠ Fig. 8.

Moats (exact numbers)

Moat	Number / claim
Scene	564 scenes / 52 buildings vs BridgeV2 24
Calibration	Extrinsics + App. G 36k / 24k refined sets
Co-train	+22% ID / +17% OOD without new algorithm
Ecosystem	Inside OXE; π_0 open mix 9.1% includes DROID; OpenPI pi05_droid ; FAST ZS campus narrative

Risks / blockers

- Embodiment lock-in (Franka+Robotiq+ZED).
- Paper policy stack is **pre-VLA** — buy *data*, rent *models* ($\pi_{0.5}$ / OpenVLA-FAST).
- Mix hostility: DROID can **hurt** underfit AR tokenizers — dip ≠ “bad data.”
- Long-horizon ceiling: Cook Lentils is a demo, not a product SLA.
- Wrist underused in paper policies despite collection.
- Crowd lang + GPT-4V scene tags need audit for mission-critical apps.
- Privacy: homes/offices RGB — check redaction before redistributing subsets.

Ops checklist (no Discord/Drive)

- Pull CC-BY + visualizer; pin **v1.0.1**-class language coverage for OpenPI paths (~75k annotated).
- Reproduce **one** Fig. 8 task at **10** rollouts before claiming domain benefit.
- VLA pretrain: per-shard action metrics + drop/downweight rules.
- BEHAVIOR: map augmentations → sim domain randomizations + real scene-rotation schedule.

Exam bite (Nov 6): “DROID shows **same-robot, many-real-scenes** teleop + protocolized diversity beats co-training on broader but shallower OXE pools for Franka robustness (**+22%/+17%**); later foundation models both **depend on** and sometimes **drop** DROID depending on capacity and action representation.”

6.2 Scientific Reviewer — WEIGHT 2 (substantial)

Claim under review: In-the-wild distributed Franka teleop with unprecedented scene diversity improves diffusion-policy success and OOD robustness vs in-domain-only and vs OXE co-training.

Strengths

1. Clear problem + **Table I** positioning (**564** scenes vs priors $\ll 100$; OXE agg ~311 but many lab-like).
2. Operationalized diversity axes (verbs, objects, scenes, viewpoints, interaction loci) — not “we collected a lot.”
3. Protocol transparency (GUI random tasks, augs, ~20 min scene cap) → replicate *process*, not only weights.

- 4. **Best causal-ish evidence = Fig. 10** (matched **7362** traj, vary scene diversity) — cite this **before** aggregate **Fig. 8** when asked what *proves* diversity.
- 5. Multi-environment eval including real household kitchen.
- 6. Honest non-goal: no new architecture; builds on Diffusion Policy correctly.
- 7. App. G calibration upgrades with quality metrics — rare for a dataset paper.

Weaknesses / actionable improvements

- 1. **Fig. 8** averages with SE; per-task tables lighter; **10** rollouts/cell thin — demand CIs + pre-registered OOD defs.
- 2. “Vs OXE” = curated Octo soup (Language Table dropped), not full 1.4M — say so.
- 3. Co-train uses first ~**40k** language-ready successes — **40k ≠ 76k**; freeze versioned manifests.
- 4. Wrist cams under-ablated despite selling point.
- 5. No zero-shot new-scene with **0** in-domain demos (authors admit open) — punish “download and it works.”
- 6. Scene dedup heuristic (App. D); GPT-4V taxonomy needs human audit rates.
- 7. Language IAA stats thin.
- 8. Long-horizon still short-chunk replay — no hierarchical planner / options baseline.
- 9. OXE confound: embodiment **match** to Franka eval may drive gains — demand Franka-only OXE slice.
- 10. Fig. 10 matches N traj but top-20 scenes may differ in task/object entropy / collector skill — match verb/object histograms.
- 11. Couples data claims to one DP head — cross-check Octo / VLA / π_0 .
- 12. Labor/safety externalities of in-the-wild collection underspecified for replication ethics.

Verdict: Accept as landmark dataset paper. Scene-diversity ablation is the scientific highlight. Overclaim risk = marketing “generalist policies” when experiments are **per-task co-trained specialists**. OpenVLA drop = **interaction effect**, not refutation.

Reviewer exam bite: Cite **Fig. 10** before **Fig. 8** to prove diversity; cite **40k vs 76k** for what was optimized; cite **Fig. 6** only for interaction-location coverage (not main SR).

6.3 Empiricist

Hypotheses (from DELIVERABLE)

- 1. **H-droid-lift:** 50/50 DROID-subset co-train > none on ID and especially OOD (Fig. 8 moral).
- 2. **H-scene>volume:** At fixed N, high scene-count subsample > top-scenes subsample on OOD (Fig. 10).
- 3. **H-vs-oxe:** On Franka-like proxies, DROID scene diversity can beat heterogeneous OXE of similar/larger N (test carefully; OXE may still win cross-embodiment).
- 4. **H-chunk-horizon:** pred **16** / exec **8** @ ~15 Hz improves multi-stage completion vs single-step BC.
- 5. **H-fail-exclude / H-lang-multi / H-protocol:** falsifiable variants in DELIVERABLE §3.

Minimal matrix: {mix: none | droid_diverse_N | droid_top_scenes_N | oxe_subset} × {init: scratch_DP_small | octo | openvla-lora} × {FT: tiny in-domain} + one chunk {8,16} contrast.

Pri	Experiment	Smoke	GPU-h
P0	Shard inventory + schema smoke	Y	<1
P0	Co-train mix card none / DROID / OXE	Y	8–24
P0	Scene-diversity matched-N (Fig. 10)	Y	10–30
P1	Horizon/chunk {8,16,32}×{4,8}; fail include; lang multi; cam-aug	Y/Partial	6–24
P1	Same mix → Octo / OpenVLA-LoRA / π_0	Partial	20–60
Defer / N	Full 76k DP × multi-site Franka; Cook Lentils household parity; 0-demo ZS	N	≫200

Success bar: loader + ≥ 2 scene mix cards with **directional** SR/MSE + GPU/disk/scene-count logs — **not** Fig. 8 household parity.

Tattoo: 76k / 350h / 564 / 86 / 52 / 50 / 2-16-8 / +22/+17 / 40k train / 7362 ablate.

Env outlines: /workspace/droid-quiet/env_outline/ (see §7).

6.4 Archaeologist

```

Tool-home teleop (DobbE, Visual Imitation Made Easy)
Bridge / BridgeV2 (WidowX scene scaling)
RH20T / RoboSet / RT-1 (lab-scale teleop)
↓
DROID (2024): same Franka, many buildings, protocolized diversity
↓
OXE aggregation (DROID as a slice; ~311 scenes pooled)
↓
Octo (curated OXE diffusion GRP) / RT-X
↓
OpenVLA: DROID @10% → DROP mid-train (action-token underfit);
      Franka-DROID Wipe FT @15 Hz wins vs DP/Octo
↓
 $\pi_0$ : open 9.1% = OXE Magic Soup + Bridge + DROID
 $\pi_0$ -FAST: DROID ZS language generalist (campus rooms)
 $\pi_{0.5}$  / OpenPI: pi05_droid full FT (RLDS v1.0.1  $\approx$  75k lang eps)
↓
BEHAVIOR'25-'26: stage-rich household; steal diversity protocol (TENTATIVE as named baseline)

```

Intellectual move: Flip OXE's "many embodiments" axis to **many authentic scenes on one research-standard arm**, with **calibration + three views** baked in.

What later papers learned

- Positive: scene diversity = first-class generalization lever.
- Negative: diverse $\neq$ easy to fit under discrete AR tokens (OpenVLA drop).
- Systems: open Franka stack = shared eval substrate for ZS claims (FAST).

Exam bite: "DROID is the **in-the-wild Franka** ancestor in the OXE $\rightarrow$ VLA $\rightarrow \pi$ chain; remember both the **co-train gains** and the **drop/underfit** sequel."

6.5 Visionary — BEHAVIOR + follow-ups + apps (no Discord/Drive)

BEHAVIOR Challenge connection

- BEHAVIOR needs policies that survive **new houses / layouts / long horizons**. DROID lesson: **maximize scene & viewpoint entropy under a fixed embodiment** with **protocolized anti-bias collection** — not only scale demos on easiest tasks.
- Map GUI ideas $\rightarrow$ BEHAVIOR ops: random BDDL task sampling per scene, forced relocation across challenge scenes, distractor/lighting jitter, keep fails for recovery research.
- Map **Fig. 10** $\rightarrow$ challenge ablations: matched-demo **many-scene vs few-scene** before claiming "needed more hours."
- Pair DROID-style real diversity priors with BEHAVIOR sim demos (co-train / filt / DR bridges).
- Motion: use **16/8** as literacy baseline; ablate vs hierarchical / $\pi_{0.5}$ -style on long BDDL; race on **$\pi_{0.5}$ / N1.7**, not 2024 ResNet DP.

One narrative line: OXE = **who** the robot is (embodiment soup); DROID = **where** it works (scene soup); BEHAVIOR 2026 = **what long-horizon household progress** looks like when both lessons meet stage structure.

Follow-up research

1. Zero-shot new scene without in-domain demos.
2. Best fusion DROID + OXE + web/VLM (avoid naive concat).
3. When does more scene tail hurt?
4. Wrist-centric / 3D-calibrated policies using App. G.
5. Action-representation × DROID fit (diffusion / FM / FAST vs naïve bins).
6. Publish a **BEHAVIOR diversity card** (layout families, distractors, cam jitter, instance splits).

New applications

- Household Franka assistants: open DROID prior + hour-scale personalization.
- Cross-campus mobile-manip shared eval rooms (FAST-style ZS).
- 3D-aware manip / Gaussian scenes from stereo+extrinsics.
- Data-quality tooling: calib QA, success detectors, lang relabel.
- Education: reproducible “second robot” teaching co-training.

Visionary exam bite: “For BEHAVIOR, steal DROID’s **scene-churn protocol** and **diversity-ablation discipline**; steal π -series **models** — don’t ship 2024 ResNet diffusion as your 2026 entry.”

7. Empiricist run pack (from DELIVERABLE)

Env setup

```
export DROID_WORK_ROOT=$HOME/src
export DROID_DATA_ROOT=$SCRATCH/droid_subsets # EDIT: large disk – subsets only
cd /workspace/droid-quiet # or copy env_outline to cluster
bash env_outline/setup_mamba.sh
# mamba activate droid-scout
mkdir -p logs
# EDIT partition/account in slurm_*.sh
sbatch env_outline/slurm_shard_smoke.sh
sbatch env_outline/slurm_cotrain_mix.sh
sbatch env_outline/slurm_scene_diversity.sh
sbatch env_outline/slurm_horizon_chunk.sh
```

File	Role
env_outline/conda_env.yaml	mamba sketch
setup_mamba.sh	env create + subset download pointers
slurm_shard_smoke.sh	1-shard schema inventory
slurm_cotrain_mix.sh	none / DROID / OXE mix cards
slurm_scene_diversity.sh	diverse-N vs top-scenes-N
slurm_horizon_chunk.sh	pred/exec sweep
run_checklist.md	GPU/disk honesty preflight

Outlines only — not sbatch’d from quiet box. Prefer official / HF resized mirrors; full stereo RGB+depth = **TB-class** — never mirror entire corpus on a login node.

GPU / RAM / disk honesty

Workload	VRAM	Disk	Wall	Smoke
Shard schema smoke	0–8 GB	20–200 GB selective	min–hours I/O	Y
Small DP co-train scout (≤10k + tiny ID)	12–24 GB	+ ckpt	4–12 h	Y
Scene-diversity pair	12–24 GB	shared	1–2 days	Y
OpenVLA-7B LoRA + mix	24–48 GB	+15–30 GB ckpt	8–24 h	Partial
Chunk sweep ×3	12–24 GB	shared	1–2 days array	Y
Full DROID mirror / paper multi-site A/B	—	≫1–several TB + robots	—	N

Honesty rule: if #SBATCH --gres=gpu:N but no device → **fail job**; log peak VRAM + **mix card hash** + **scene cardinality** next to every SR. Never claim “full 76k” from a capped download.

Success criteria vs paper

Paper claim	Feasible bar	Not required
Usable in-the-wild Franka data	Loader streams multi-cam / action / lang / success	Full 76k ingest
Co-train ↑ robustness	DROID-subset > none on one OOD proxy	+22/+17 across 6 real tasks
Scene diversity drives OOD	diverse-N > top-scenes-N @ matched N	Exact Fig. 10 bars
Chunk helps long-horizon	Chunked ↑ stage completion on one multi-stage proxy	Household Cook Lentils robot

Pass: mix cards + ≥1 directional **H-droid-lift** or **H-scene>volume** with logs.
Fail / overclaim: “we reproduced Fig. 8 household results” without robots / full protocol.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Stakeholder	5 · longest	Product, moats, midterm decisions, OXE contrast
Reviewer	2 · substantial	Fig. 10 before Fig. 8; 40k vs 76k; landmark + punish
Empiricist	filled	Mix cards + scene-N + chunk smoke + GPU honesty
Archaeologist	filled	OXE → DROID → OpenVLA drop → π_0 /FAST/pi05_droid → BEHAVIOR
Visionary	filled	Steal protocol into BEHAVIOR; no Discord/Drive

Spine (paste into Slide Maker):

- Same Franka, many real scenes** (564 / 52 buildings) — opposite of OXE’s many-robots axis.
- Protocol > vibes:** GUI random tasks, ~20 min scene churn, view/lighting augs.
- Fig. 8 +22% ID / +17% OOD; Fig. 10** proves diversity at matched 7.3k — cite Fig. 10 before Fig. 8 for causality.
- Motion:** chunked DP **obs=2 / pred=16 / action=8 @15 Hz** — reactive BC, **not** a classical planner; Cook Lentils / Clean Desk = long-horizon stress.
- BEHAVIOR steal:** diversity cards + scene-churn protocol onto $\pi_{0.5}$ /N1.7 — don’t retrain paper DP or just drink another river; midterm Nov 6 needs these numbers.

GitHub title sketch: [ECE 605] - DROID Making Stakeholder (or Reviewer if seat assigned)

Paths

Artifact	Path
This pack	/workspace/droid-quiet/KANTA_PACK.md
PDF	/workspace/droid-quiet/KANTA_PACK.pdf
Senior Research	/workspace/droid-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/droid-2403.12945-summary.md
Ablations	/workspace/papers/droid-2403.12945-ablations.md
Env outlines	/workspace/droid-quiet/env_outline/
Paper PDF	/workspace/papers/droid-2403.12945.pdf

Pack synthesized 2026-09-05 PT for KanBot / ECE 605 Paper Manager — Nov 4 Data · Stakeholder=5 · Reviewer=2 · midterm Nov 6 nearby.